

What's past is prologue: FRAP keeps delivering 50 years later

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ABSTRACT Fluorescence recovery after photobleaching (FRAP) has emerged as one of the most widely utilized techniques to quantify binding and diffusion kinetics of biomolecules in biophysics. Since its inception in the mid-1970s, FRAP has been used to address an enormous array of questions including the characteristic features of lipid rafts, how cells regulate the viscosity of their cytoplasm, and the dynamics of biomolecules inside condensates formed by liquid-liquid phase separation. In this perspective, I briefly summarize the history of the field and discuss why FRAP has proven to be so incredibly versatile and popular. Next, I provide an overview of the extensive body of knowledge that has emerged on best practices for quantitative FRAP data analysis, followed by some recent examples of biological lessons learned using this powerful approach. Finally, I touch on new directions and opportunities for biophysicists to contribute to the continued development of this still-relevant research tool.

SIGNIFICANCE FRAP has become a household word in the biophysics and cell biology communities. Here, I explore what makes FRAP such a powerful technique.

INTRODUCTION

Perhaps no technique in biophysics has provided as many insights into the molecular motions that underpin life as fluorescence recovery after photobleaching (FRAP). FRAP was first conceived in the 1970s as a tool to study the lateral mobility of proteins at the cell surface (1–5) (reviewed in (6,7)) (Fig. 1). The enormous potential of FRAP as a tool to provide broader insights into the dynamics of cellular components, including cytosolic and cytoskeletal proteins, became apparent early on (8–10). However, membrane-centric applications of FRAP continued to dominate the field for many years (11,12).

Thanks to the development of GFP technology and concomitant advances in fluorescence microscopy, FRAP underwent a resurgence 20 years after its inception (7,13,14). Early applications of confocal FRAP reshaped cell biology and biophysics by revealing the highly dynamic nature of protein and lipid dynamics in intracellular compartments. They also helped to uncover organelle connectivity and the kinetics and directionality of intracellular

trafficking pathways in unprecedented detail (7,13,14). Applications of FRAP subsequently exploded to address questions as wide ranging as the nature of lipid rafts (15,16), mechanisms controlling nucleocytoplasmic transport (17–19), reaction-diffusion kinetics governing intracellular binding events (20,21), timescales of protein turnover in muscle (22,23), mass transport and diffusion within bacterial biofilms (24,25), impact of macromolecular crowding on diffusion (26–29), transcriptional kinetics (30–33), protein synthesis rates (34–37), kinetics of receptor recycling (38,39), and the evolution of morphogen gradients (40–42). FRAP has also expanded beyond biophysics and cell biology, proving its utility in pharmaceutical applications (43), colloid science (44), material science and industrial applications (45,46), and even in food science (47–49).

Even today, the applications of FRAP are continuing to grow. Here, I summarize the features of FRAP that have contributed to its immense staying power and discuss some of the newest applications and biophysical lessons learned from this remarkably robust technique.

WHAT MAKES FRAP SO VERSATILE?

The principle of FRAP is deceptively simple (Figs. 1 and 2). A biomolecule of interest within a cell or model system is

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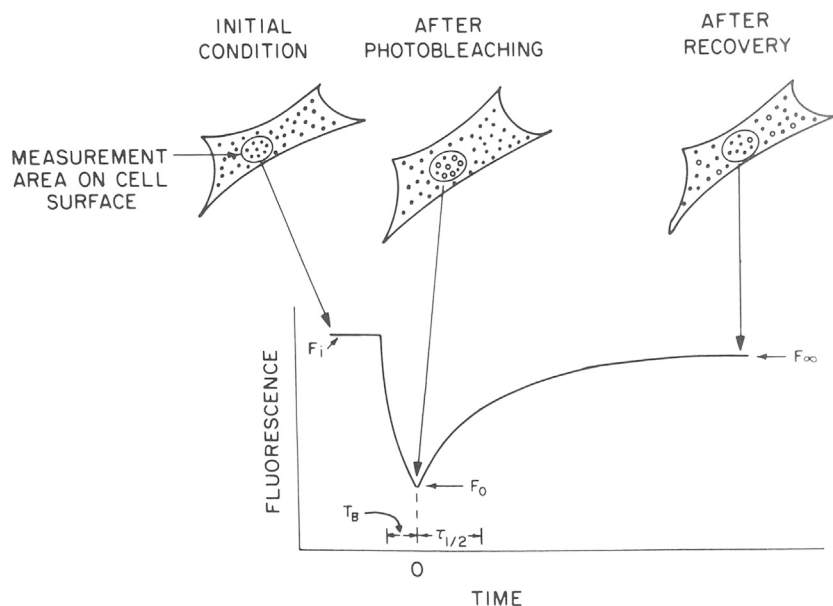


FIGURE 1 Early cartoon depicting the principle of FRAP measurements of the lateral diffusion of cell surface components as illustrated in a 1976 paper from Jacobson and colleagues (3). In this cartoon, the area subjected to photobleaching and used to monitor fluorescence recovery is marked by the oval on the cell. F_i , fluorescence intensity prior to photobleaching; F_0 , fluorescence intensity immediately following photobleaching; F_∞ , fluorescence intensity after recovery is complete. τ_B corresponds to the time required for the photobleaching, and $\tau_{1/2}$ is the characteristic half time of recovery. Reproduced with permission from John Wiley and Sons, *J. Cell. Biochem.*, Feb 19, 2004, G. Poste, Y. Hou, E-S. Wu, et al., "Measurement of the lateral mobility of cell surface components in single living cells by fluorescence recovery after photobleaching." Copyright 1976 Alan R. Liss, Inc.

fluorescently labeled, enabling its detection using fluorescence microscopy. A subpopulation of the fluorescent molecules is then irreversibly photobleached using a transient pulse of high-intensity excitation within a region of interest (ROI). Exchange of fluorescent molecules from the still-fluorescent surrounding regions with those in the dark bleached region is then monitored over time. Next, an FRAP curve is obtained by plotting the values of fluorescence intensity in the selected ROI.

FRAP curves report on two types of information. The first is the kinetics of recovery, reflecting the movement of fluorescent molecules from the surrounding area into the bleached ROI. This is empirically characterized by the time required to reach 50% of complete recovery. The kinetics of recovery depend not only on the properties of the system under study but also on the size and shape of the bleach region. The second general parameter contained within FRAP curves is the fraction of molecules that exhibit recovery over the time course of the experiment, commonly referred to as the mobile fraction.

Two major mechanisms typically determine the extent and kinetics of recovery in FRAP experiments: diffusion (i.e., Brownian motion) and reaction (in this context, corresponding to binding events). In general, diffusion depends on the size and shape of the diffusing species, temperature, and the viscosity of the medium through which it diffuses. It also depends on whether obstacles or barriers to diffusion exist. FRAP is also sensitive to how the diffusing species interacts or binds to other molecules or structural features found in its environment during its journey. FRAP thus can be used to study not only the properties of the diffusing species itself but also probe its environment. These factors can be understood/dissected through a combination of

experimental approaches—for example by systematically varying the diffusing species or its environment—together with applications of appropriate models that capture the underlying biophysical mechanisms to fit the FRAP curves. Relevant examples include free diffusion, anomalous diffusion, and reaction-diffusion models (5,20,21,40,50–61).

Several features of FRAP make it particularly versatile compared with other biophysical methods used to study the motions of biomolecules in cells such as fluorescence correlation spectroscopy and single-particle tracking. Although most commonly applied to study the dynamics of GFP-tagged proteins, FRAP measurements are amenable to any class of molecules that can be fluorescently labeled (15,21,45,46,62–68). FRAP can be coupled to a variety of microscopy modalities including wide field, confocal, multiphoton, TIRF, light sheet, and even superresolution microscopy (7,52,69–75). As discussed further below, the bleaching step can be performed using a wide range of bleach ROI sizes, geometries, and patterns (5,63,70,76–83) (Fig. 2). FRAP can thus be used to study motions in structures and environments across length scales varying in size from diffraction-limited spots to populations of cells.

FRAP is also well suited to resolve recoveries that occur over a range of timescales. Although typically employed to monitor transport processes with characteristic recoveries of seconds to minutes (Fig. 2), FRAP can even be used to follow recoveries that occur over hours. Examples of recovery processes that take place over these significantly longer time frames include myosin incorporation into myofibers and the dynamics of titin assembly and disassembly in sarcomeres (22,23,84,85).

Finally, FRAP can be easily carried out in almost any system that can be studied using fluorescence microscopy.

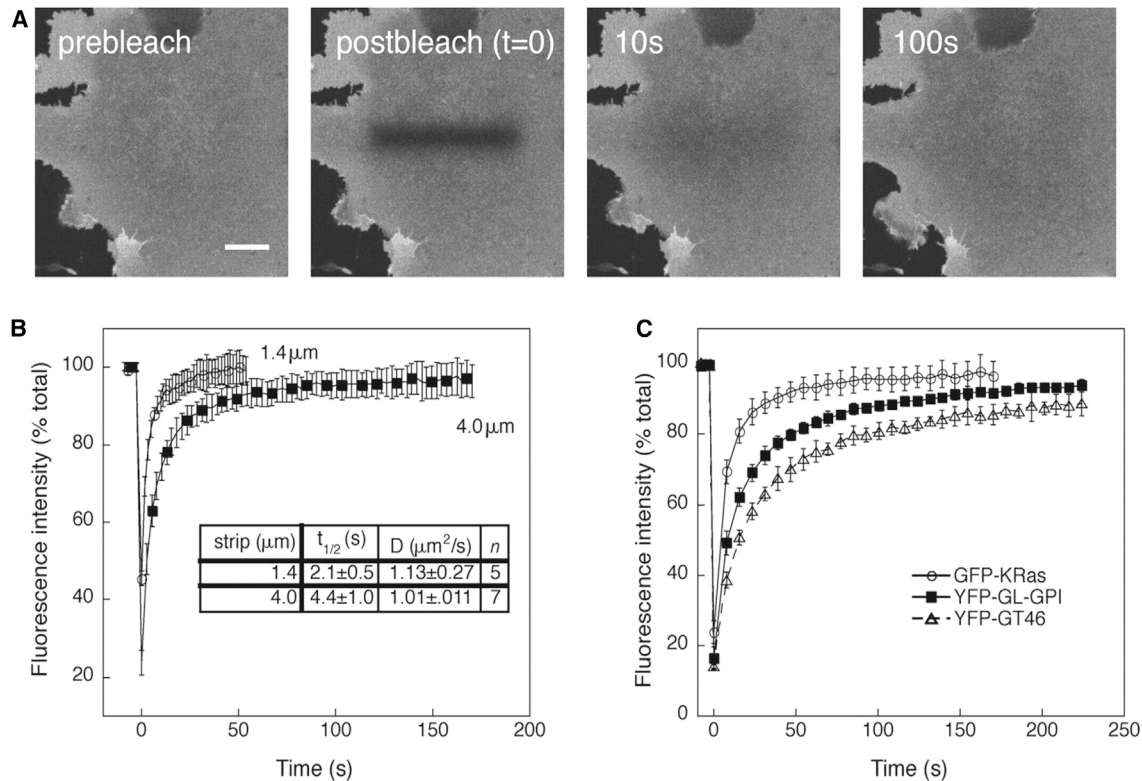


FIGURE 2 Example of confocal FRAP analysis of cell surface proteins (16). (A) Selected images from a confocal FRAP experiment in cells expressing a GFP-tagged plasma membrane protein (GFP-KRas), using a rectangular bleach region 4 μm wide. Note that only a portion of the cell is shown in this image. Bar, 10 μm. (B) Comparison of FRAP results obtained using bleach boxes of different sizes for cells expressing GFP-KRas. (C) Kinetics of recovery differ for three different cell surface proteins. FRAP experiments were performed using identically sized bleach boxes (4 μm wide). Used with permission of Rockefeller University Press, from *J. Cell Biol.*, Volume 165, A. K. Kenworthy et al., "Dynamics of putative raft-associated proteins at the cell surface," pp. 735–46, Copyright 2004. Permission conveyed through Copyright Clearance Center, Inc.

In addition to its widespread use in in vitro models and cultured cells, FRAP has been employed in diverse organisms such as sea urchins (8,86), *E. coli* (87–91), *C. elegans* (92–96), zebrafish (41,97–99), and *Drosophila* (68,100–105). In other recent incarnations, FRAP is even being used to study motions of biomolecules in mice using intravital imaging (106–108).

In summary, FRAP is an exceptionally flexible method. This flexibility is an important hallmark of FRAP that underlies its widespread utility. However, it also demands careful attention be paid to design of FRAP experiments as well as data analysis, for reasons discussed further below.

TO HARNESS THE FULL POWER OF FRAP, DETAILS MATTER

Although the principle of FRAP is straightforward, the details of FRAP experiments and analysis matter immensely; a one-size-fits-all approach cannot be applied (43,51,109–112). One example is the choice of bleach region. Depending on the question being addressed and the instrumentation being used to perform the measurement, bleach regions can range in size as well as shape, including diffraction-limited

spots, circular discs, lines, strips, and even more complex geometries (5,70,76–82). It can also be advantageous to deliberately vary the size or shape of the bleach region to probe for specific recovery processes. This is especially useful to test for recoveries that are governed primarily by binding kinetics, which unlike diffusion are independent of the size of the bleach region (20,50). This approach has been used to study the binding kinetics of transcription factors in the nucleus as well as parse out contributions of lateral diffusion and membrane-cytosolic exchange to the dynamics of peripheral membrane proteins (20,50,93,113–115).

The choice of the best FRAP model to analyze the data is critical. FRAP recoveries can be fit using simple exponential equations to extract characteristic halftimes or time constants of recoveries and calculate mobile/immobile fractions (51,116). However, this approach does not take advantage of the ability of FRAP to yield quantitative measurements of diffusion coefficients and kinetic constants. To do so, FRAP models must account for the geometry and size of the bleach region (5,51,70,76–82). Other important considerations for FRAP models include boundary conditions, whether recovery occurs in one, two, or three dimensions,

and the shape of the initial bleach profile (51,52,79,117–123). Analytical models have been derived for some limiting cases (reviewed in (45)). However, numerical methods, simulations, and even deep learning approaches have proven valuable, for example in the analysis of FRAP experiments using arbitrary bleach region shapes or to take three-dimensional cellular geometries and recoveries into account (76,117,124–128). The importance of choosing the correct FRAP model and validating its underlying assumptions has been periodically revisited as new applications of FRAP emerge (50,51,55,64,79,109,112,117,129–131). An example of how these factors influence quantitative insights into the dynamics of phase-separated condensates is discussed below.

Rigorous analysis of FRAP data also requires attention to potential caveats of the technique. Underlying assumptions in FRAP experiments include 1) the photobleaching event is irreversible, 2) changes in fluorescence observed during the recovery solely reflect the dynamics of the fluorescently labeled molecules, and 3) the fluorescence labeling and bleaching strategies are nonperturbing. This is not always the case (66,75,105,132–139). For example, some fluorescent proteins undergo photoswitching events, thus voiding the assumption that photobleaching is irreversible (75,132–135). This effect can interfere with the accurate interpretation of FRAP data, but it can be overcome by building it into the model used to fit the data or using fluorescent probes that do not exhibit photoswitching behavior (64,75,134,135). Conversely, potential contributions of new protein synthesis to fluorescence recovery must be considered for experiments monitoring dynamics of genetically encoded fluorescent proteins that occur over long time-scales (23,34,36,37,135). Thus, even when the underlying assumptions of FRAP do not hold, such issues can often be corrected for.

RECENT EXAMPLES OF FRAP IN ACTION

Through its rich history, FRAP has already uncovered an enormous body of knowledge on the mechanisms that govern the motions of biomolecules in cells. Yet even today it is continuing to provide new biological and biophysical insights. One recent example comes from studies of the regulation of intracellular viscosity, a topic first investigated using FRAP in the 1980s (9). It has been known for many years that the viscosity of cell membranes of organisms that do not control their body temperature adapts to environmental conditions through changes in lipid composition, a process known as homoviscous adaptation (86,140–143). Maintenance of cytoplasmic viscosity is equally important, because it is a rate limiting factor in determining how rapidly molecules can find one another in the cytoplasm. Very recently, it was discovered that budding yeast buffer their cytoplasmic viscosity by regulating the synthesis of two carbohydrates, trehalose and glycogen, in response to

changes in temperature and nutrient availability (144,145). Key evidence in support of this viscoadaptation model came from FRAP experiments of the reporter protein GFP in vitro in solutions of varying viscosity, in cell lysates, and in the cytoplasm of living cells (144).

FRAP is also yielding new insights into how cells modulate nucleocytoplasmic transport kinetics in response to mechanical stimuli. Cargoes shuttle between the nucleus and cytoplasm by passing through nuclear pores via either passive or facilitated diffusion. The utility of FRAP to monitor these processes has been recognized for many years (17,18). In these experiments, FRAP is typically used to photobleach fluorescent molecules of interest in either the nucleus or cytoplasm and then to monitor their exchange between the two compartments over time. It has long been recognized that the nuclear pore complex restricts passive transport of shuttling cargo in a size-dependent manner (146). With the help of FRAP, it has now become clear that the permeability of the nuclear pore complex changes when mechanical force is applied to the nucleus (147–149). This in turn has important implications for how the entry and exit of transcriptional regulators into and out of the nucleus is regulated (147,148).

CONDENSATES EMERGE AS THE LATEST TARGET OF FRAP ANALYSIS

The extensive knowledge base that has developed over the years makes lessons learned from FRAP readily transferable to new biological phenomena. A prime example is biomolecular condensates. Formation of these membraneless organelles has now emerged as a key organizing principle in cell biology and biophysics (150). Condensates are dynamic structures that form as a consequence of liquid-like phase separation driven by weak multivalent interactions. The diffusivity of fluorescently labeled molecules within condensates is readily assessed using FRAP measurements of subregions within individual condensates in vitro and in vivo (Fig. 3) (51,109,151–155). Furthermore, components can exchange between condensates and their surroundings. This can be appreciated from experiments in which entire condensates are completely photobleached. Depending on the properties of the condensates and their environment, the kinetics and features of these two classes of recoveries can be quite different from one another (51).

The pivotal role of FRAP as a tool to study condensates is readily apparent from the rapid proliferation of methods articles in this area (51,130,156–164). A number of studies have applied simple exponential fits or a simplified 2D model assuming an infinite boundary to analyze the resulting data (summarized in (51)). However, condensates represent somewhat of a special case for FRAP analysis (Fig. 3). They exist as three-dimensional droplets and are relatively small compared with the size of bleach regions typically employed in FRAP experiments. Moreover, they exchange

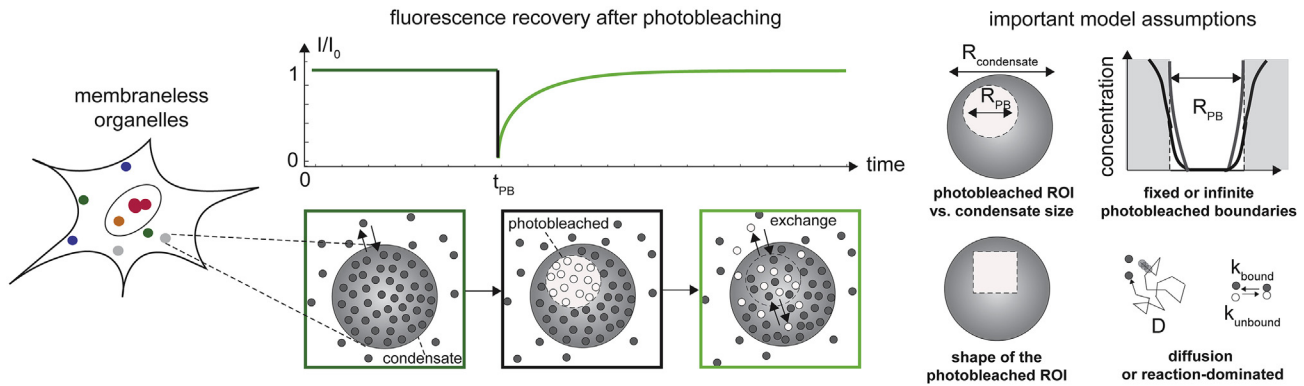


FIGURE 3 Considerations for quantitative FRAP measurements of membraneless organelles. FRAP of subregions of individual biomolecular condensates yields information about the mobility of molecules within the condensate as well the kinetics of exchange of molecules into and out of the condensate. Factors that need to be considered to model the resulting FRAP curves correctly include the ratio of the size of the photobleached region versus condensate size, the shape of the photobleached region, and whether recoveries are reaction- or diffusion-dominated. Reprinted with permission from Elsevier from *Biophys. J.* Volume 177, A. Soranno, “The trap in the FRAP: a cautionary tale about transport measurements in biomolecular condensates.” pp 2041–42. Copyright 2019. To see this figure in color, go online.

components with their environment, and the concentration and diffusional mobility of their components can vary inside and outside of the droplets. Given these factors, which assumptions and models to analyze FRAP experiments of condensates are most valid? To address this question, Taylor and colleagues systematically compared the results obtained using different FRAP models to fit FRAP measurements of reconstituted condensates in vitro as well as intracellular condensates (51). Importantly, they found that depending on the model used, the resulting diffusion coefficients varied by an order of magnitude. This work not only highlights the importance of the correct choice of FRAP models to study condensates per se, but it also provides an excellent overview of the many factors that must be considered to extract quantitative measurements in complex systems from a best practice perspective using FRAP.

OUTLOOK FOR THE FUTURE

All indications suggest that FRAP is here to stay and poised to continue to make critical contributions in new arenas of biology and biophysics. Many opportunities exist for biophysicists to continue to contribute to this ever-burgeoning field. One example is the development of new FRAP probes. Although GFP variants remain a major workhorse for FRAP studies of protein dynamics, the development of new fluorescent proteins and alternative approaches to fluorescently label molecules of interest potentiate new applications of FRAP (68,103). Opportunities also exist to continue to expand the footprint of FRAP to new imaging modalities, each with its own unique set of challenges to consider. Examples of some of the most recent couplings include light sheet microscopy (71–73,165), superresolution microscopy (74), and intravital imaging (106–108,166). As FRAP continues to expand to new areas of biology and biophysics,

biophysicists will continue to play an important role in developing and validating new FRAP models to enable quantitative measurements, as well as dissecting the biophysical principles governing the behavior of each system (51,109,130). Efforts to develop new approaches to visualize and analyze transport of biomolecules within systems of increasing complexity are already underway (73,167). Finally, to support researchers from other fields, it will also be increasingly important to continue to disseminate best practices and find new ways to democratize FRAP analysis (63,70,79,124,167–171). Thanks to the founders of FRAP, even 50 years later many exciting new discoveries still lie ahead.

AUTHOR CONTRIBUTIONS

A.K.K. conceived the topic, wrote the original version of the manuscript, selected the figures, and edited the manuscript.

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DECLARATION OF INTERESTS

The author declares no competing interests.

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